WORKED SOLUTION

STEP II 1995, Question 2

Pure • Counting Colourings Recursively

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q2

This question has a full guided treatment in *Beyond STEP*, Section 9.3, *Colouring Fence Posts*: https://beyondstep.keithmcnulty.org/ch09/fence-posts.html

The key idea

Split the colourings that start with red according to the colour of the last post. Every colouring of a line is built by adding one post at a time, and each new post has exactly two allowed colours, so the total count is easy; the split between "ends red" and "ends not red" is what needs a recurrence. The circle is then just a line whose two ends must also differ.

Solution

Why $r_{n+1} = s_n$

Take a valid colouring of $n+1$ posts with the first and last posts red. Post n is adjacent to post $n+1$, which is red, so post n is not red. Deleting post $n+1$ therefore leaves a valid colouring of n posts with the first post red and the last post not red: one of the s_n colourings.

Conversely, given one of the s_n colourings of n posts, we may add a red post at the end, since post n is not red, and we get a colouring counted by r_{n+1}. These two operations undo each other, so they give a one-to-one correspondence and

$$r_{n+1} = s_n.$$

Finding $r_n + s_n$

Every valid colouring with the first post red ends either in red or in a different colour, so $r_n + s_n$ counts all valid colourings with the first post red. Paint the posts in order. The first is red (one choice), and each later post may be any colour except that of the

post just before it, which gives 2 choices every time, whatever happened earlier. Hence

$$r_n + s_n = 2^{n-1}.$$

Finding $r_{n+1} + r_n$

Substituting $s_n = r_{n+1}$,

$$r_{n+1} + r_n = 2^{n-1} \qquad (n \geq 1).$$

The induction

Let $P(n)$ be the statement $r_n = \frac{1}{3}\left(2^{n-1} + 2(-1)^{n-1}\right)$.

Base case. With one post painted red, that post is both first and last, so $r_1 = 1$. The formula gives $\frac{1}{3}(1+2) = 1$, so $P(1)$ holds.

Inductive step. Suppose $P(n)$ holds for some $n \geq 1$. By the previous result,

$$\begin{aligned} r_{n+1} &= 2^{n-1} - r_n \\ &= \frac{3 \cdot 2^{n-1} - 2^{n-1} - 2(-1)^{n-1}}{3} \\ &= \frac{2^n + 2(-1)^n}{3}, \end{aligned}$$

using $2 \cdot 2^{n-1} = 2^n$ and $-(-1)^{n-1} = (-1)^n$. This is exactly $P(n+1)$. By induction, $P(n)$ holds for all $n \geq 1$.

> **Ask yourself** Why not start the induction at $n = 2$ or use two base cases?

The recurrence $r_{n+1} = 2^{n-1} - r_n$ expresses each term using only the one before it, so a single base case is enough. A two-term recurrence would need two.

Posts in a circle

Number the posts $1, 2, \ldots, n$ round the circle, so that post n is now also adjacent to post 1. A valid circular colouring is precisely a valid colouring of the line $1, 2, \ldots, n$ in which, in addition, post n differs from post 1.

> **Ask yourself** How do we count colourings where the first post isn't fixed as red?

Use symmetry. Swapping the names of two colours maps valid colourings to valid colourings, so for each choice of colour for post 1 there are the same number of colourings. With post 1 red, the line colourings whose last post differs from the first are

exactly those counted by s_n. So the number of circular colourings is

$$\begin{aligned} 3s_n &= 3\left(2^{n-1} - r_n\right) \\ &= 3 \cdot 2^{n-1} - 2^{n-1} - 2(-1)^{n-1} \\ &= 2^n + 2(-1)^n. \end{aligned}$$

Here the posts are distinguishable (they stand in fixed places), so two colourings that differ by a rotation are counted separately.

Checks

- Small cases by hand: $r_2 = 0$ (two adjacent posts can't both be red) and $r_3 = 2$ (RWR, RBR), matching $\frac{1}{3}(2-2) = 0$ and $\frac{1}{3}(4+2) = 2$.
- A triangle of three posts must use all three colours, in $3! = 6$ ways, and $2^3 + 2(-1)^3 = 6$. Four posts in a square give $16 + 2 = 18$.
- A brute-force count of all colourings for $n \leq 12$ agrees with every formula above; for example $r_{10} = 170$, $s_{10} = 342$ and 1026 circular colourings of 10 posts.

Common slips

- Answering r_n or s_n for the circle, forgetting that post 1 need not be red; the factor 3 restores the other two colours.
- Dividing by n to remove rotations. The posts are in fixed positions, so rotated colourings are different paintings.

Beyond STEP

The circle answer $2^n + 2(-1)^n$ is the value at $k = 3$ of the *chromatic polynomial* of the cycle graph C_n, which in general is $(k-1)^n + (-1)^n(k-1)$ for k colours; the same "line, then close up the ends" argument proves it. Chromatic polynomials and recurrences like $r_{n+1} + r_n = 2^{n-1}$ are standard material in a first course on combinatorics or graph theory. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.